AIM-Fair: Advancing Algorithmic Fairness via Selectively Fine-Tuning Biased Models with Contextual Synthetic Data

Taro Yamada

Abstract

*Models trained on real-world datasets often inherit demographic biases, leading to disparate performance across protected groups. Existing approaches to mitigating such bias through generative data augmentation struggle with low synthetic data quality, domain gaps between synthetic and real distributions, and a reliance on costly demographic annotations. We present **AIM-Fair**, a framework that advances algorithmic fairness through two synergistic components: Contextual Synthetic Data Generation (CSDG) and Selective Fine-Tuning (SFT). CSDG harnesses GPT-4 to produce rich, diverse textual prompts describing facial characteristics (*e.g.*, facial features, head pose, lighting) that guide a Stable Diffusion model to generate demographically balanced, high-quality images—requiring no group labels on real data. SFT then identifies which model parameters to update via a Selective Mask Generation (SMG) procedure: gradient differences across three datasets (biased real, biased synthetic, and unbiased synthetic) are used to rank parameters by fairness sensitivity and domain sensitivity, and only parameters that are both fairness-sensitive and domain-insensitive are updated. This selective strategy improves worst-group accuracy while resisting the domain shift introduced by synthetic data. Extensive experiments on CelebA and UTKFace demonstrate that AIM-Fair achieves state-of-the-art fairness-utility trade-offs, surpassing prior generative methods including DiGA across multiple target and protected attribute settings, and remaining robust under data-scarce conditions.*

1. Introduction

Facial attribute classification (FAC) is a fundamental task in computer vision with broad real-world applications, including surveillance, human-computer interaction, and medical imaging [8, 18]. Despite significant advances driven by large-scale datasets and deep neural networks [11], FAC models trained on real-world data routinely exhibit troubling demographic disparities: a model may be highly accurate overall while performing substantially worse for specific subgroups defined by protected attributes such as gender, age, or race [19, 28].

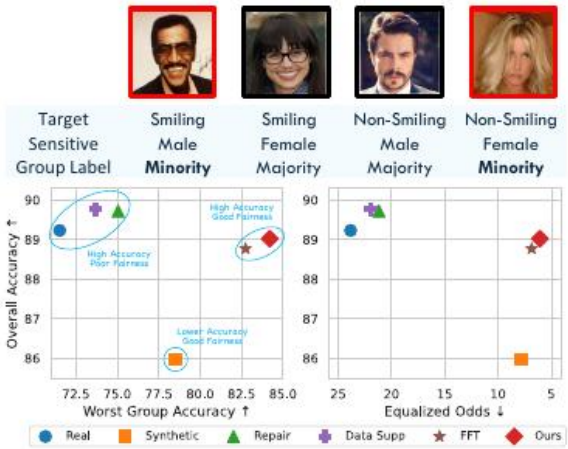

Figure 1. Facial attribute classification (FAC) on the CelebA dataset based on different training strategies, where ***Smiling*** is the target attribute and ***Gender*** is the protected attribute. AIM-Fair substantially improves worst-group accuracy while maintaining competitive overall accuracy.

These disparities arise from biases embedded in training data—spurious statistical correlations between target attributes (*e.g.*, *Smiling*) and protected attributes (*e.g.*, *Male*) that the model learns to exploit, producing unfair predictions [21, 30]. For example, as illustrated in Fig. 1, a classifier trained to predict *Smiling* on a gender-biased dataset achieves high overall accuracy but performs poorly on minority demographic groups (*e.g.*, non-smiling males or smiling females), reflecting a model that has learned gender as a spurious cue rather than the target attribute alone.

A broad spectrum of debiasing strategies has been proposed, broadly categorized as pre-processing (data rebalancing), in-processing (training objective modification), and post-processing methods [3, 9, 19]. Distributionally robust optimization (DRO) approaches [7, 10, 16] seek models that perform well under worst-case data distributions, while invariant learning methods [1, 4] attempt to learn features that are invariant to spurious correlations. However,

many of these approaches require group-annotated training data—an assumption that is often violated in practice due to the cost and privacy concerns of collecting demographic labels.

Recent generative data augmentation methods offer a promising label-free alternative by synthesizing additional training samples to rebalance demographic groups [5, 23, 26, 30]. The leading approach, DiGA [30], identifies spurious attributes via generative models and augments training with edited images. While effective, generative augmentation methods face two core challenges: (1) *data quality and diversity*: synthetically generated images may lack the visual richness and diversity of real data, and (2) *domain shift*: fine-tuning on out-of-distribution synthetic data can degrade overall accuracy by overwriting valuable knowledge encoded in the pre-trained model.

In this paper, we introduce **AIM-Fair** (**A**dvancing Algorith**m**ic **Fair**ness), a framework that addresses both challenges through two novel components. First, our **Contextual Synthetic Data Generation (CSDG)** module uses GPT-4 to automatically produce rich, context-aware textual descriptions of facial images—capturing attributes such as skin tone, hairstyle, expression, and head pose—that guide Stable Diffusion [25] to generate high-quality, balanced images across demographic groups without any real-data group annotations. Second, our **Selective Fine-Tuning (SFT)** module uses a novel **Selective Mask Generation (SMG)** procedure to identify model parameters that are both fairness-sensitive and domain-insensitive. By computing gradient differences across biased real, biased synthetic, and unbiased synthetic datasets, SMG ranks parameters according to two criteria: how much each parameter contributes to the fairness gap between biased and unbiased distributions, and how susceptible each parameter is to domain shift between real and synthetic data. Only the intersection of top-k fairness-sensitive and domain-insensitive parameters is updated during fine-tuning, preserving overall accuracy while improving worst-group performance.

The main contributions of this work are:

- We propose **AIM-Fair**, a fine-tuning paradigm that improves fairness using synthetic data without requiring demographic group annotations of real data.
- We introduce **CSDG**, which leverages GPT-4 contextual prompts combined with Stable Diffusion to generate diverse, high-quality, bias-controlled training images.
- We develop **SMG**, a parameter-wise selective fine-tuning strategy based on gradient-difference ranking to isolate fairness-sensitive, domain-insensitive model parameters.
- We demonstrate state-of-the-art fairness-utility trade-offs on CelebA [18] and UTKFace [31], outperforming full fine-tuning, partial fine-tuning, and prior generative methods including DiGA [30].

2. Related Work

Algorithmic Fairness. Algorithmic fairness has attracted substantial attention as machine learning systems are deployed in high-stakes domains [3, 19]. Fairness criteria include demographic parity [6], equalized odds [9], and group-accuracy-based metrics [10]. In the vision domain, biases in facial attribute classification are well-documented [8, 21, 28], with performance disparities observed across gender, age, and race subgroups.

Fairness via Robust Training. Distributionally robust optimization (DRO) methods train models to minimize worst-case risk over a family of data distributions [7, 10, 16]. CVaR DRO [16] minimizes the conditional value-at-risk to improve group fairness. JTT [17] upweights samples misclassified by an initial ERM model to improve robustness on minority groups. EIIL [4] infers environment partitions and applies invariant risk minimization [1]. LfF [20] employs a generalized cross-entropy objective to learn debiased features. MAPLE [32] exploits model prediction disagreement for fairness-aware training. While effective, many such methods require sensitive attribute labels during training, limiting their applicability.

Generative Augmentation for Fairness. A complementary line of work synthesizes data to rebalance training distributions [2, 23, 26, 27]. Ramaswamy *et al*. [23] use GANs for counterfactual data generation. Shin *et al*. [26] augment underrepresented groups by filling the data distribution gap. D'Incà *et al*. [5] and Ye *et al*. [29] use vision-language models and diffusion-based editing to augment images for improved fairness. DiGA [30] identifies spurious attributes via generative models and trains fair classifiers through distribution-generative augmentation—representing the current state of the art. Our work differs from DiGA in two key aspects: (1) we generate fully synthetic images via context-rich prompts rather than editing real images, and (2) we use selective fine-tuning to protect domain knowledge encoded in the pre-trained model.

Selective and Parameter-Efficient Fine-Tuning. Selective fine-tuning methods update a subset of model parameters to improve task performance while preserving pre-trained knowledge [12, 14, 34]. Sub-network tuning [14] selects individual layers or blocks based on task-specific criteria. LoRA [13] introduces low-rank updates to model weights. Lee *et al*. [15] propose surgical fine-tuning guided by gradient statistics. In contrast, AIM-Fair performs *parameter-wise* selection using gradient differences across multiple datasets, enabling fine-grained control over which parameters capture fairness-relevant vs. domain-relevant information.

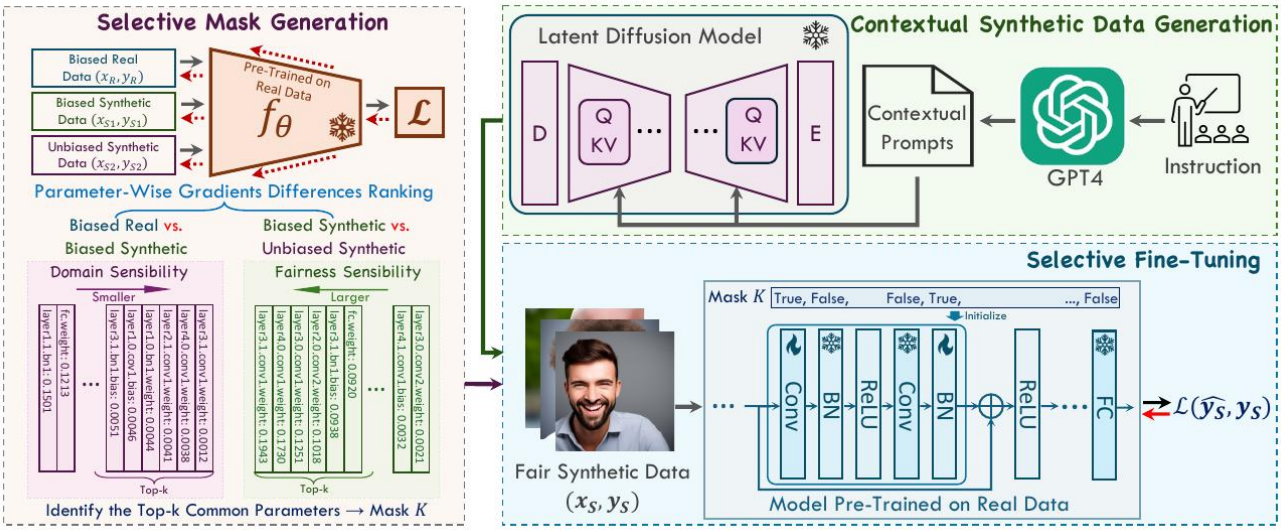

Figure 2. Overview of AIM-Fair. The framework consists of three components: (1) **Contextual Synthetic Data Generation (CSDG)**: GPT-4 generates diverse, context-aware prompts that guide Stable Diffusion to produce balanced synthetic images across demographic groups. (2) **Selective Mask Generation (SMG)**: gradient differences across three datasets (biased real $\mathcal{D}_R$, biased synthetic $\mathcal{D}_{S_1}$, and unbiased synthetic $\mathcal{D}_{S_2}$) are used to rank parameters by fairness sensitivity ($|\mathbf{g}_{S_1} - \mathbf{g}_{S_2}|$) and domain sensitivity ($|\mathbf{g}_R - \mathbf{g}_{S_1}|$), producing a binary selection mask. (3) **Selective Fine-Tuning (SFT)**: only the masked parameters are updated using balanced synthetic data.

Text-to-Image Generation. Latent diffusion models [25] have transformed text-conditioned image synthesis, enabling high-quality, photorealistic generation from natural language descriptions. Building on CLIP [22], models like Stable Diffusion [25] and DALL-E [24] can generate images with fine-grained attribute control. Recent work has explored using large language models to generate diverse prompts for downstream vision tasks [33]. Our CSDG module builds on this line of work, using GPT-4-generated prompts to guide Stable Diffusion toward balanced demographic synthesis for fairness.

3. Method

AIM-Fair improves the fairness of a biased pre-trained classifier through a two-phase pipeline, as illustrated in Fig. 2. In Phase 1, CSDG generates a large, demographically balanced synthetic dataset without any real-data group annotations. In Phase 2, SMG identifies fairness-sensitive and domain-insensitive parameters from the pre-trained model, and SFT updates only those parameters on the balanced synthetic data.

3.1. Problem Formulation

Let $f_\theta : \mathcal{X} \to \mathcal{Y}$ be a classifier parameterized by θ, pre-trained on a biased real dataset $\mathcal{D}_R = \{(x_i, y_i, a_i)\}$ where x_i is an image, $y_i \in \{0, 1\}$ is the target attribute label, and $a_i \in \{0, 1\}$ is the protected attribute label. The dataset is biased in the sense that y and a are spuriously correlated: e.g., in CelebA the *Smiling* attribute is disproportionately associated with the *Female* subgroup. Our goal is to fine-tune f_θ to achieve group fairness—specifically, to minimize the *equalized odds* (EO) [9] gap:

$$\text{EO} = \frac{1}{2} \sum_{y \in \{0,1\}} \left| P(\hat{y} = 1 \mid Y = y, A = 0) - P(\hat{y} = 1 \mid Y = y, A = 1) \right|, \tag{1}$$

without significantly degrading overall accuracy, and *without* access to protected attribute labels on the real training data.

3.2. Contextual Synthetic Data Generation (CSDG)

A key challenge in generative augmentation for fairness is producing synthetic images that are (a) visually realistic, (b) demographically balanced, and (c) contextually diverse, so that the model does not overfit to unrealistically homogeneous synthetic images. We address this through CSDG, which uses GPT-4 as a *contextual prompt generator*.

GPT-4 Prompt Generation. Given a target attribute $y \in \{0, 1\}$ and protected attribute $a \in \{0, 1\}$ (*e.g.*, $y = 1$ = Smiling, $a = 0$ = Female), GPT-4 is queried to generate N diverse textual descriptions of human faces exhibiting the given combination of y and a. These descriptions go beyond simple attribute labels to include contextual details such as:

- Facial features: age appearance, skin tone, hairstyle, makeup

Figure 3. Generated images of *Smiling Male* conditioned on different GPT-4-generated contextual prompts with head pose variations, demonstrating high visual diversity and realism across samples.

- Expression nuances: slight smile, broad grin, teeth visible
- Head pose: frontal, 15° left turn, slight tilt
- Lighting and background context

This contextual richness is crucial. As shown in Tab. 5 and Fig. 3, a plain prompt such as "a smiling woman" produces images with low intra-group diversity and poor fine-tuning performance, whereas contextual prompts from GPT-4 yield substantially improved fairness metrics. Adding head pose variation further improves worst-group performance.

Stable Diffusion Generation. Each GPT-4-generated prompt p is used to condition Stable Diffusion v2.1 [25] to generate a set of images. We generate an equal number of images for each of the four demographic-target combinations $(y, a) \in \{0, 1\}^2$, producing a balanced synthetic dataset $\mathcal{D}_{S_2}$ with approximately equal representation across groups. We also generate a biased synthetic dataset $\mathcal{D}_{S_1}$ with the same group imbalance as $\mathcal{D}_R$, used for SMG (described below).

By default, we use 50 contextual prompts per group (including head pose variations) and generate synthetic data matching the real training set size (ratio 1:1). Fig. 3 illustrates the visual diversity achieved by CSDG compared to plain prompting.

3.3. Selective Mask Generation (SMG)

Directly fine-tuning all parameters of f_θ on synthetic data risks two failure modes: (1) overwriting domain-specific knowledge learned from real data, causing accuracy degradation; and (2) updating parameters insensitive to the fairness gap, wasting capacity on irrelevant adjustments. SMG identifies the subset of parameters that are *fairness-sensitive* and *domain-insensitive*.

Gradient Computation. Let $\mathbf{g}_R$, $\mathbf{g}_{S_1}$, $\mathbf{g}_{S_2}$ denote the mean gradient vectors computed over datasets $\mathcal{D}_R$, $\mathcal{D}_{S_1}$, $\mathcal{D}_{S_2}$ respectively, at the pre-trained checkpoint θ^*:

$$\mathbf{g}_{\mathcal{D}} = \frac{1}{|\mathcal{B}|} \sum_{B \in \mathcal{B}} \nabla_\theta \mathcal{L}(f_{\theta^*}, B), \tag{2}$$

where $\mathcal{B}$ is the set of mini-batches from dataset $\mathcal{D}$, and $\mathcal{L}$ is binary cross-entropy loss.

Fairness Sensitivity Score. The fairness sensitivity of parameter θ_j measures how differently the model responds to biased versus unbiased data:

$$s_j^{\text{fair}} = \left| \mathbf{g}_{S_1}^{(j)} - \mathbf{g}_{S_2}^{(j)} \right|. \tag{3}$$

A high fairness sensitivity score indicates that parameter θ_j is strongly affected by the fairness gap between biased and balanced distributions—meaning updating it could improve group fairness.

Domain Sensitivity Score. The domain sensitivity of θ_j measures the discrepancy between gradients on real versus synthetic data with the same bias:

$$s_j^{\text{dom}} = \left| \mathbf{g}_R^{(j)} - \mathbf{g}_{S_1}^{(j)} \right|. \tag{4}$$

A high domain sensitivity score indicates that θ_j is strongly domain-dependent—updating it on synthetic data would cause domain shift and degrade real-data performance.

Selective Mask. We define the selective mask $\mathcal{M}$ as the set of parameters that rank in the top-k by fairness sensitivity *and* in the bottom-k by domain sensitivity (i.e., domain-insensitive):

$$\mathcal{M} = \text{Top-}k(s^{\text{fair}}) \cap \text{Bottom-}k(s^{\text{dom}}). \tag{5}$$

This intersection ensures that fine-tuning improves fairness (high s^{fair}) without degrading accuracy due to domain shift (low s^{dom}). The parameter k controls the sparsity of fine-tuning.

3.4. Selective Fine-Tuning (SFT)

Given the mask $\mathcal{M}$, SFT fine-tunes only the selected parameters on the balanced synthetic dataset $\mathcal{D}_{S_2}$:

$$\theta^{\text{fair}} = \arg \min_{\theta : \theta_j \text{ fixed for } j \notin \mathcal{M}} \mathcal{L}(f_\theta, \mathcal{D}_{S_2}). \tag{6}$$

Algorithm 1 AIM-Fair

Require: Biased real dataset $\mathcal{D}_R$; pre-trained model f_{θ^*}; top-k budget

// Phase 1: Contextual Synthetic Data Generation
Generate contextual prompts via GPT-4 for each $(y, a) \in \{0, 1\}^2$
Generate balanced synthetic dataset $\mathcal{D}_{S_2}$ using Stable Diffusion
Generate biased synthetic dataset $\mathcal{D}_{S_1}$ using same prompts with group imbalance
// Phase 2: Selective Fine-Tuning
Compute mean gradients $\mathbf{g}_R, \mathbf{g}_{S_1}, \mathbf{g}_{S_2}$ at θ^* via back-propagation
Compute fairness scores $s_j^{\text{fair}} = |\mathbf{g}_{S_1}^{(j)} - \mathbf{g}_{S_2}^{(j)}|$ for all j
Compute domain scores $s_j^{\text{dom}} = |\mathbf{g}_R^{(j)} - \mathbf{g}_{S_1}^{(j)}|$ for all j
Compute mask $\mathcal{M} = \text{Top-}k(s^{\text{fair}}) \cap \text{Bottom-}k(s^{\text{dom}})$
Freeze θ_j for all $j \notin \mathcal{M}$
Fine-tune f_θ on $\mathcal{D}_{S_2}$ to obtain θ^{fair}
return Fair classifier $f_{\theta^{\text{fair}}}$

All other parameters θ_j with $j \notin \mathcal{M}$ are frozen during fine-tuning. This selective update preserves the model's general representational capacity learned from real data while adjusting only the fairness-relevant parameters.

The full procedure is summarized in Algorithm 1.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate on two standard facial attribute classification benchmarks:

- **CelebA** [18]: Large-scale face attribute dataset with over 200K images and 40 binary attribute labels. We use *Smiling* and *Young* as target attributes (T) and *Male* and *Young* as protected attributes (P), following the protocol of DiGA [30].
- **UTKFace** [31]: Dataset with over 20K face images annotated with age, gender, and race. We use *Female* as the target attribute and *White* as the protected attribute.

For CelebA, training data is constructed with a bias ratio of 1:9 (minority:majority group), following prior work [17, 30].

Metrics. We report three evaluation metrics: (1) **ACC**: overall classification accuracy; (2) **WST**: worst-group accuracy (minimum across the four demographic-label subgroups); (3) **EO**: equalized odds gap (Eq. (1)). Higher ACC and WST indicate better performance; lower EO indicates better fairness.

Baselines. We compare against ERM [11], CVaR DRO [16], EIIL [4], LfF [20], JTT [17], MAPLE [32], and DiGA [30]. We additionally compare to training strategies including training solely on synthetic data, data supplementation [26], data repairing [5, 29], linear probing, and full fine-tuning. For selective parameter strategies, we compare against best random selection and Sub-Tuning [14].

Implementation Details. For the pre-trained classifier, we use ResNet-18 [11] pre-trained on ImageNet. Pre-training uses SGD with learning rate 0.01, momentum 0.9, and weight decay 10^{-4} for 15 epochs. Selective fine-tuning uses SGD with learning rate 0.05 for 10 epochs. Images are resized to 224×224. The default top-k value is $k = 40\%$ of all parameters. Experiments are averaged over multiple random seeds. For CSDG, we use GPT-4 to generate 50 contextual prompts per group (combining facial feature descriptions with head pose variation) and generate 5,000 images per (y, a) group using Stable Diffusion v2.1.

4.2. Comparison with State-of-the-Art Methods

Tab. 1 compares AIM-Fair with baseline methods on CelebA across three attribute combinations, and with data-scarce variants. On **T: Smiling, P: Male**, AIM-Fair achieves ACC 89.02, WST 84.20, and EO 6.07, surpassing the best prior method DiGA (ACC 88.40, WST 81.90, EO 7.40) across all three metrics. On **T: Smiling, P: Young**, AIM-Fair achieves ACC 90.21, WST 87.78, EO 5.44, markedly outperforming DiGA (ACC 89.10, WST 78.50, EO 9.50)—a substantial 9.28 point improvement in WST. On **T: Young, P: Male**, AIM-Fair achieves WST 54.89 and EO 28.18, improving both fairness metrics over DiGA (WST 51.30, EO 33.30), with slightly lower ACC (78.19 vs. 80.00). Notably, AIM-Fair consistently achieves better WST and EO than all baselines while remaining competitive in overall accuracy, demonstrating an improved fairness-utility trade-off.

Under **data-scarce settings** (T: Smiling, P: Male), AIM-Fair remains robust when real training data is reduced. At 50% of real data, it achieves ACC 88.85, WST 83.99, EO 6.28; at 25%, ACC 88.89, WST 82.54, EO 7.59; at 10%, ACC 87.90, WST 81.73, EO 8.16—consistently outperforming DiGA and other baselines. This robustness is particularly important as it demonstrates that AIM-Fair's synthetic augmentation can compensate for limited real data.

4.3. Ablation: Training Strategy

Tab. 3 compares different training strategies for applying synthetic data. Training solely on synthetic data achieves the best EO (3.50 for T: Smiling, P: Young) but at the cost of substantially lower ACC (85.12 vs. 90.21), due to the domain gap between synthetic and real test data. Data supplementation [26] and data repairing [5, 29] maintain

Table 1. Comparisons to other methods on the CelebA dataset under settings of varied target and protected attributes.

Methods	T: Smiling, P: Male			T: Smiling, P: Young			T: Young, P: Male		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [11]	88.20	70.10	25.30	88.30	71.50	15.60	77.70	42.00	52.00
CVaR DRO [16]	87.30	74.00	22.80	87.00	76.10	13.90	75.40	42.30	48.80
EIIL [4]	87.90	75.60	19.70	87.90	72.50	13.30	77.50	45.60	39.20
LfF [20]	87.10	77.50	17.00	85.30	72.90	14.30	77.40	44.20	43.60
JTT [17]	88.00	74.80	19.40	87.60	73.30	14.20	76.30	43.60	47.70
MAPLE [32]	88.10	72.00	19.60	88.10	73.60	13.60	76.30	46.20	43.50
DiGA [30]	88.40	81.90	7.40	89.10	78.50	9.50	**80.00**	51.30	33.30
AIM-Fair (Ours)	**89.02**	**84.20**	**6.07**	**90.21**	**87.78**	**5.44**	78.19	**54.89**	**28.18**

Table 2. Comparisons to other methods on the CelebA dataset (T=Smiling, P=Male) under settings of training set sizes.

Methods	Samples Ratio = 50%			Samples Ratio = 25%			Samples Ratio = 10%		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [11]	87.50	67.80	26.10	87.10	65.90	27.70	86.90	62.80	28.90
CVaR DRO [16]	86.60	72.90	22.10	86.60	72.30	22.40	85.50	69.10	27.30
EIIL [4]	86.20	71.30	22.50	85.90	69.60	25.40	86.80	64.20	26.70
LfF [20]	86.90	75.50	19.40	85.90	72.10	23.60	85.50	66.10	27.70
JTT [17]	87.30	72.90	20.10	86.70	71.10	20.60	86.80	67.10	23.10
MAPLE [32]	87.40	73.70	23.80	87.00	72.70	24.20	85.60	69.20	27.10
DiGA [30]	88.40	81.10	7.80	88.40	78.30	8.00	**88.30**	78.80	8.40
AIM-Fair (Ours)	**88.85**	**83.99**	**6.28**	**88.89**	**82.54**	**7.59**	87.90	**81.73**	**8.16**

Table 3. Comparisons of varied training strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA: Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				UTKFace: Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Baseline [11]	T=0	83.57	98.50	89.23	71.52	23.84	10.65	78.72	96.09	90.19	78.72	17.37	6.92	79.58	96.16	88.86	79.58	16.59	7.26
	T=1	95.36	71.52					94.37	86.42					95.75	83.96				
Trained on Synthetic Data	T=0	86.82	89.40	85.98	78.52	7.87	4.07	85.45	85.76	85.12	82.42	**3.50**	**1.34**	82.71	88.12	85.44	82.71	5.50	**1.96**
	T=1	86.39	78.52					82.42	85.27					84.90	86.04				
Data Supplementation [26]	T=0	84.39	98.39	**89.77**	73.67	21.96	9.76	78.05	95.36	90.35	78.05	17.31	7.02	82.12	93.97	**90.25**	82.12	11.85	5.24
	T=1	95.40	73.67					94.90	87.62					95.62	89.28				
Data Repairing [5, 29]	T=0	82.82	98.10	89.72	75.05	21.18	9.50	79.50	95.19	**90.68**	79.50	15.69	6.30	81.15	94.22	89.42	81.15	13.37	5.89
	T=1	96.04	75.05					94.51	88.35					95.71	86.61				
Linear Prob	T=0	84.55	96.22	89.69	77.41	17.51	7.71	85.20	93.81	90.14	85.20	8.61	3.18	81.06	91.51	88.40	81.06	10.59	5.14
	T=1	94.82	77.41					90.43	87.88					94.59	86.42				
Fully Fine-Tuning	T=0	88.81	91.62	88.77	82.74	6.83	3.31	85.98	90.06	87.85	85.98	4.37	1.68	83.00	88.04	87.90	83.00	5.40	2.99
	T=1	89.55	82.74					86.02	86.81					90.74	89.83				
AIM-Fair (Ours)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	**4.81**	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

high ACC but do not substantially improve over the baseline ERM model in terms of fairness. Linear probing improves fairness somewhat but leaves accuracy unchanged compared to full fine-tuning. Full fine-tuning on synthetic data achieves competitive WST and EO, but AIM-Fair's selective strategy surpasses it in both WST (+1.46 on Smiling/Male) and EO (-0.76 on Smiling/Male) while maintaining similar overall accuracy, confirming the benefit of selective parameter updating.

4.4. Ablation: Selective Fine-Tuning Strategy

Tab. 4 evaluates different partial fine-tuning strategies. Random selection of k parameters achieves reasonable results but is outperformed by principled gradient-based strategies. Sub-Tuning [14] by updating one block achieves the highest ACC (90.04 on Smiling/Male) but with worse fairness metrics (WST 80.64, EO 12.45). Sub-Tuning by freezing one block improves EO but still lags behind AIM-Fair in WST. A cosine-similarity-based selection criterion shows similar trends. AIM-Fair (absolute gradient difference) achieves the best WST (84.20) and EO (6.07) on Smiling/Male—demonstrating that the absolute gradient difference is a more effective criterion than cosine similarity or block-wise selection for isolating fairness-sensitive, domain-insensitive parameters.

4.5. Ablation: Top-k Parameter Budget

Fig. 4 shows the effect of varying the top-k percentage of parameters updated during SFT. With too few parameters updated ($k \to 0$), the model cannot improve fairness. With too many ($k \to 100\%$), SFT approaches full fine-tuning and

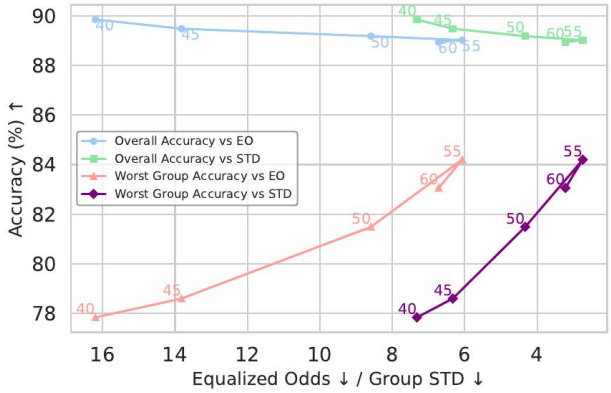

Figure 4. Effect of the top-k parameter budget on CelebA (T=Smiling, P=Male). Setting $k = 40\%$ achieves optimal trade-off between ACC, WST, and EO.

domain shift degrades performance. We find that $k = 40\%$ achieves the best balance, yielding optimal WST and EO while maintaining competitive ACC. This result validates the importance of careful parameter selection.

4.6. Ablation: Layer-Wise Analysis

To further understand which model components benefit most from selective fine-tuning, we conduct a layer-wise analysis on CelebA (T=Smiling, P=Male). Fig. 5 shows results from freezing individual layers of ResNet-18. Freezing only the fully connected (FC) layer already improves WST and reduces EO relative to the biased baseline, sug-

Table 4. Comparisons of varied partial fine-tuning strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA												UTKFace							
		Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Best Random Selection	T=0	87.93	92.81	89.33	82.17	9.14	4.09	85.52	91.53	88.95	85.52	6.02	2.16	83.50	88.85	87.92	83.50	5.38	2.61
	T=1	91.31	82.17					88.25	87.63					90.30	89.01				
Best Sub-Tuning [14] (By Updating One Block)	T=0	87.50	94.75	**90.04**	80.64	12.45	5.52	87.28	94.63	**90.33**	87.28	7.34	3.00	83.05	89.56	**89.01**	83.05	6.51	3.56
	T=1	93.09	80.64					89.15	87.32					92.07	91.34				
Best Sub-Tuning [14] (By Freezing One Block)	T=0	88.69	91.09	89.06	83.48	7.00	3.00	86.55	91.89	88.65	86.55	**5.34**	2.25	83.58	88.34	87.42	83.58	**4.76**	**2.23**
	T=1	90.48	83.48					87.08	86.55					88.96	88.81				
Selective Fine-Tuning (Cosine Similarity)	T=0	86.07	91.77	88.85	83.33	8.40	3.61	87.43	92.96	90.26	87.43	5.53	**2.05**	83.72	88.82	88.21	83.72	5.10	2.64
	T=1	91.53	83.33					89.52	88.65					90.14	90.15				
AIM-Fair (Ours) (Absolute Difference)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	4.81	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

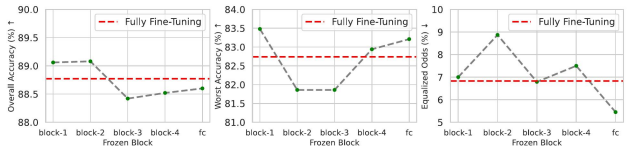

Figure 5. Effects of selective fine-tuning by layer-wise freezing on CelebA (T=Smiling, P=Male). Freezing only the FC layer improves worst-group accuracy and reduces equalized odds.

Table 5. Classification results on CelebA dataset (T=Smiling, P=Male) under settings of different prompt types and numbers.

Prompt Types (Number)	Targe	Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Plain Prompt (1)	t=0	65.18	89.44	71.65	43.28	34.20	17.07	83.10	77.72	72.75	59.53	9.30	8.96
	t=1	77.47	43.28					59.53	68.83				
Contextual Prompts (25)	t=0	77.12	89.25	81.85	69.73	16.3	7.65	76.72	67.90	79.80	67.90	8.82	8.75
	t=1	85.99	69.73					85.16	91.09				
Contextual Prompts (50)	t=0	89.47	89.80	82.66	72.22	**5.07**	7.66	87.72	90.43	**85.85**	79.67	4.36	4.20
	t=1	77.26	72.22					79.67	82.67				
Contextual + Head Poses (50)	t=0	86.82	89.40	**85.98**	**78.52**	7.87	**4.07**	85.45	85.76	85.12	**82.42**	**3.50**	**1.34**
	t=1	86.39	78.52					82.42	85.27				

gesting that the FC layer is particularly domain-sensitive. Selectively updating only convolutional layers in the middle of the network tends to be most beneficial for fairness improvement. AIM-Fair's parameter-wise approach naturally discovers and exploits this structure without requiring manual layer selection.

4.7. Ablation: Prompt Design in CSDG

Tab. 5 studies how prompt design affects downstream fairness. Using a single plain prompt (*e.g.*, "a smiling woman") results in poor performance due to low intra-group diversity and limited coverage of the real test distribution. Increasing to 25 contextual prompts substantially improves WST from 43.28 to 69.73 on Smiling/Male, with further gains at 50 prompts. Crucially, adding head pose variation to contextual prompts provides the best WST (78.52) and STD (4.07) on Smiling/Male, and achieves the best WST (82.42) and STD (1.34) on Smiling/Young. This demonstrates that head pose diversity is important for generating a well-distributed synthetic dataset.

Table 6. Classification results on CelebA dataset (T=Smiling, P=Male) with different number of synthetic data.

Ratio To Real Data	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
0.5	t=0	88.65	93.96	**89.51**	80.80	10.31	4.90
	t=1	91.10	80.80				
1	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				
1.5	t=0	89.18	92.05	88.34	81.25	7.13	3.98
	t=1	88.38	81.25				
2	t=0	89.01	91.26	88.81	83.36	6.08	2.96
	t=1	91.26	89.44				

4.8. Ablation: Amount of Synthetic Data

Tab. 6 examines the effect of synthetic data quantity. We vary the ratio of synthetic to real training samples from 0.5 to 2. The optimal setting (ratio=1) achieves the best WST (84.20) and EO (6.07). Both too little and too much synthetic data degrade performance, suggesting a saturation effect where excess synthetic images introduce redundancy or distribution mismatch. This confirms that quality and balance of synthetic data matter more than sheer quantity.

4.9. Ablation: Bias Ratio of Synthetic Data

Tab. 7 investigates how the bias ratio of $\mathcal{D}_{S_1}$ (the imbalanced synthetic dataset used in SMG) affects the quality of the selective mask. We vary the imbalance from mild (4:6 minority-to-majority ratio) to extreme (1:9). Increasing the bias ratio improves fairness: at 1:9, AIM-Fair achieves WST 84.20 and EO 6.07, the best results across all ratios. This is because a more biased $\mathcal{D}_{S_1}$ provides a stronger contrast signal for identifying fairness-sensitive parameters.

5. Conclusion

We have presented AIM-Fair, a framework for improving fairness in facial attribute classification through contextual synthetic data generation and selective parameter fine-tuning. AIM-Fair addresses two fundamental challenges of generative augmentation for fairness: data quality and diversity, and domain shift during fine-tuning. Our CSDG module uses GPT-4 to generate rich, diverse prompts that guide Stable Diffusion to produce demographically bal-

Table 7. Classification results on CelebA dataset (T=Smiling, P=Male) with varied bias ratio of the biased synthetic data.

Bias Ratio	Target	Protected P=0	Protected P=1	ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
Fully Fine-Tuning	t=0	88.81	91.62	88.77	82.74	6.83	3.31
	t=1	89.55	82.74				
4 : 6	t=0	85.72	91.71	**89.23**	83.90	8.78	3.77
	t=1	92.69	83.90				
3 : 7	t=0	87.05	91.62	88.90	83.52	7.72	3.26
	t=1	90.93	83.52				
2 : 8	t=0	88.01	91.76	88.97	83.16	7.30	3.28
	t=1	90.46	83.16				
1 : 9	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				

anced synthetic images without group labels on real data. Our SMG procedure identifies model parameters that are both fairness-sensitive and domain-insensitive via gradient difference ranking, enabling SFT to selectively improve worst-group accuracy without degrading overall accuracy.

Extensive experiments on CelebA and UTKFace demonstrate that AIM-Fair achieves state-of-the-art fairness-utility trade-offs, outperforming prior methods including DiGA and a range of data augmentation and partial fine-tuning baselines. Ablation studies confirm the contribution of each component: contextual prompts with head pose diversity, gradient-difference-based parameter selection, and the optimal top-k budget all play important roles. AIM-Fair is also robust to reduced real training data, making it especially appealing in low-resource settings.

While AIM-Fair represents a significant advance, several limitations and future directions merit attention. The framework currently assumes binary protected attributes; extending to multi-valued or intersectional protected attributes is an important open problem. Generating synthetic data via diffusion models carries a computational overhead; more efficient generation pipelines could improve scalability. Finally, exploring AIM-Fair in domains beyond facial attribute classification—such as medical imaging or natural language processing—represents a promising direction for broadening its impact on algorithmic fairness.

References

[1] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019. 1, 2

[2] Kristy Choi, Aditya Grover, Trisha Singh, Rui Shu, and Stefano Ermon. Fair generative modeling via weak supervision. In *ICML*, pages 1887–1898, 2020. 2

[3] Sam Corbett-Davies, Johann D Gaebler, Hamed Nilforoshan, Ravi Shroff, and Sharad Goel. The measure and mismeasure of fairness. *JMLR*, 24(1):14730–14846, 2023. 1, 2

[4] Elliot Creager, Jörn-Henrik Jacobsen, and Richard Zemel. Environment inference for invariant learning. In *ICML*, pages 2189–2200, 2021. 1, 2, 5, 6

[5] Moreno D'Incà, Christos Tzelepis, Ioannis Patras, and Nicu Sebe. Improving fairness using vision-language driven image augmentation. In *WACV*, pages 4695–4704, 2024. 2, 5, 6

[6] Michele Donini, Luca Oneto, Shai Ben-David, John S Shawe-Taylor, and Massimiliano Pontil. Empirical risk minimization under fairness constraints. *NeurIPS*, 31, 2018. 2

[7] John C Duchi and Hongseok Namkoong. Learning models with uniform performance via distributionally robust optimization. *The Annals of Statistics*, 49(3):1378–1406, 2021. 1, 2

[8] Laura Gustafson, Chloe Rolland, Nikhila Ravi, Quentin Duval, Aaron Adcock, Cheng-Yang Fu, Melissa Hall, and Candace Ross. Facet: Fairness in computer vision evaluation benchmark. In *ICCV*, pages 20370–20382, 2023. 1, 2

[9] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. *NeurIPS*, 29, 2016. 1, 2, 3

[10] Tatsunori Hashimoto, Megha Srivastava, Hongseok Namkoong, and Percy Liang. Fairness without demographics in repeated loss minimization. In *ICLR*, pages 1929–1938, 2018. 1, 2

[11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 1, 5, 6

[12] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. In *ACL*, pages 328–339, 2018. 2

[13] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. In *ICLR*, pages 1–13, 2022. 2

[14] Gal Kaplun, Andrey Gurevich, Tal Swisa, Mazor David, Shai Shalev-Shwartz, and Eran Malach. Less is more: Selective layer finetuning with subtuning. *arXiv preprint arXiv:2302.06354*, 2023. 2, 5, 6, 7

[15] Yoonho Lee, Annie S Chen, Fahim Tajwar, Ananya Kumar, Huaxiu Yao, Percy Liang, and Chelsea Finn. Surgical fine-tuning improves adaptation to distribution shifts. In *ICLR*, 2022. 2

[16] Daniel Levy, Yair Carmon, John C Duchi, and Aaron Sidford. Large-scale methods for distributionally robust optimization. *NeurIPS*, 33:8847–8860, 2020. 1, 2, 5, 6

[17] Evan Z Liu, Behzad Haghgoo, Annie S Chen, Aditi Raghunathan, Pang Wei Koh, Shiori Sagawa, Percy Liang, and Chelsea Finn. Just train twice: Improving group robustness without training group information. In *ICML*, pages 6781–6792, 2021. 2, 5, 6

[18] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *ICCV*, 2015. 1, 2, 5

[19] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6):1–35, 2021. 1, 2

[20] Junhyun Nam, Hyuntak Cha, Sungsoo Ahn, Jaeho Lee, and Jinwoo Shin. Learning from failure: De-biasing classifier

from biased classifier. *NeurIPS*, 33:20673–20684, 2020. 2, 5, 6

[21] Sungho Park, Jewook Lee, Pilhyeon Lee, Sunhee Hwang, Dohyung Kim, and Hyeran Byun. Fair contrastive learning for facial attribute classification. In *CVPR*, pages 10389–10398, 2022. 1, 2

[22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763, 2021. 3

[23] Vikram V Ramaswamy, Sunnie SY Kim, and Olga Russakovsky. Fair attribute classification through latent space de-biasing. In *CVPR*, pages 9301–9310, 2021. 2

[24] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*, pages 1–27, 2022. 3

[25] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pages 10684–10695, 2022. 2, 3, 4

[26] Joonghyuk Shin, Minguk Kang, and Jaesik Park. Fill-up: Balancing long-tailed data with generative models. *arXiv preprint arXiv:2306.07200*, 2023. 2, 5, 6

[27] Mei Wang and Weihong Deng. Mitigating bias in face recognition using skewness-aware reinforcement learning. In *CVPR*, pages 9322–9331, 2020. 2

[28] Zeyu Wang, Klint Qinami, Ioannis Christos Karakozis, Kyle Genova, Prem Nair, Kenji Hata, and Olga Russakovsky. Towards fairness in visual recognition: Effective strategies for bias mitigation. In *CVPR*, pages 8919–8928, 2020. 1, 2

[29] Moon Ye-Bin, Nam Hyeon-Woo, Wonseok Choi, Nayeong Kim, Suha Kwak, and Tae-Hyun Oh. Exploiting synthetic data for data imbalance problems: Baselines from a data perspective. *arXiv preprint arXiv:2308.00994*, 2023. 2, 5, 6

[30] Fengda Zhang, Qianpei He, Kun Kuang, Jiashuo Liu, Long Chen, Chao Wu, Jun Xiao, and Hanwang Zhang. Distributionally generative augmentation for fair facial attribute classification. In *CVPR*, pages 22797–22808, 2024. 1, 2, 5, 6

[31] Zhifei Zhang, Yang Song, and Hairong Qi. Age progression/regression by conditional adversarial autoencoder. In *CVPR*, 2017. 2, 5

[32] Xiao Zhou, Yong Lin, Renjie Pi, Weizhong Zhang, Renzhe Xu, Peng Cui, and Tong Zhang. Model agnostic sample reweighting for out-of-distribution learning. In *ICML*, pages 27203–27221, 2022. 2, 5, 6

[33] Yongchao Zhou, Hshmat Sahak, and Jimmy Ba. Using synthetic data for data augmentation to improve classification accuracy. In *ICML*, 2023. 3

[34] Fuzhen Zhuang, Zhiyuan Qi, Keyu Duan, Dongbo Xi, Yongchun Zhu, Hengshu Zhu, Hui Xiong, and Qing He. A comprehensive survey on transfer learning. *Proceedings of the IEEE*, 109(1):43–76, 2020. 2